Unsupervised and supervised analysis of protein sequences

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Motivation

- Protein sequences are high-dimensional and discrete
- It is not clear how sequence structure relates to function
- Machine learning can be used to:
  - explore the structure of sequence space
  - identify patterns without using labels
  - test whether function can be predicted from sequence data

Outline

1. Dataset and representation
2. Exploring sequence space (unsupervised)
3. Can function be predicted? (supervised)
4. Learning a generative model
5. Validating generated sequences
6. Conclusions

Dataset and representation

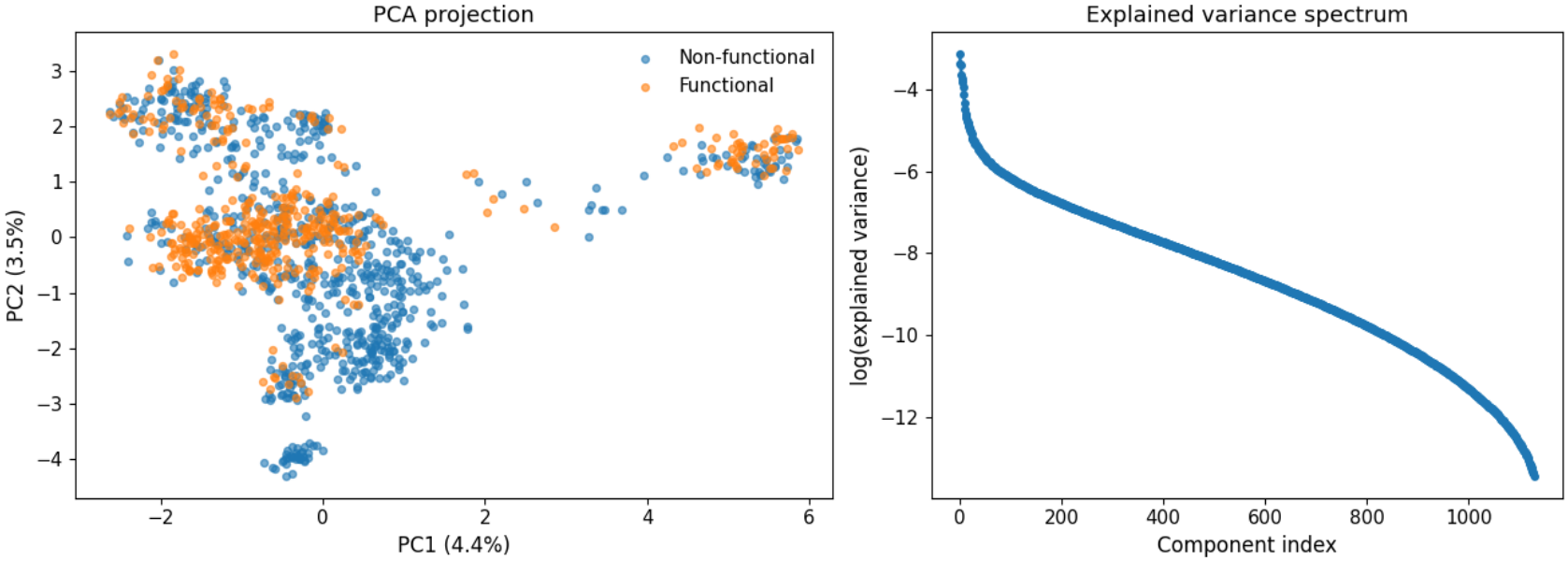
Input data

- Aligned protein sequences of fixed length L
- Alphabet of 20 amino acids (plus gap symbol)
- Each sequence is a string: $(a_1, a_2, \ldots, a_L)$

Numerical representation

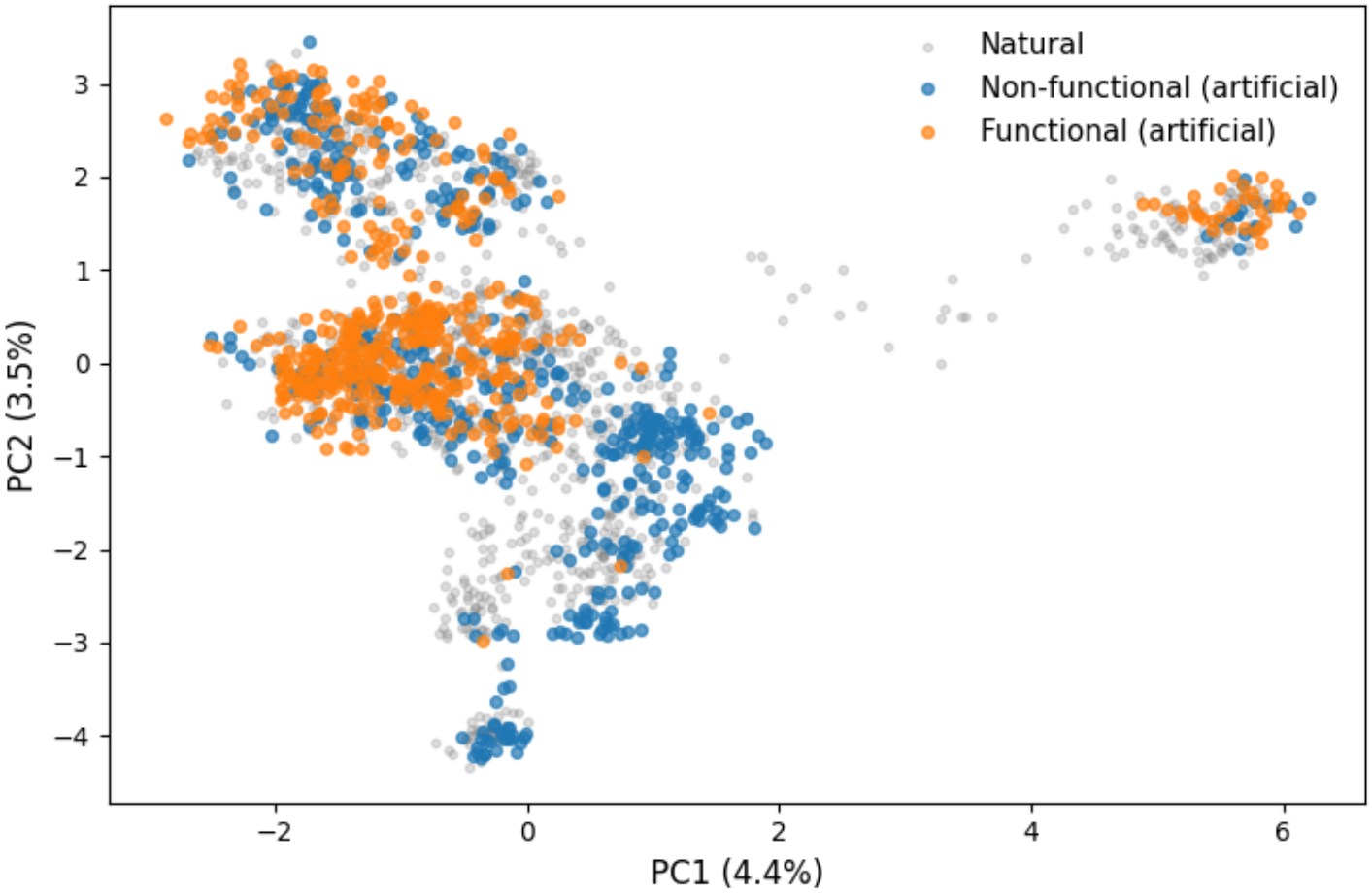
- Machine learning models require numerical inputs
- Each amino acid is encoded using one-hot representation
- Each sequence is mapped to a vector in $\mathbb{R}^{20L}$

Exploring sequence space: PCA



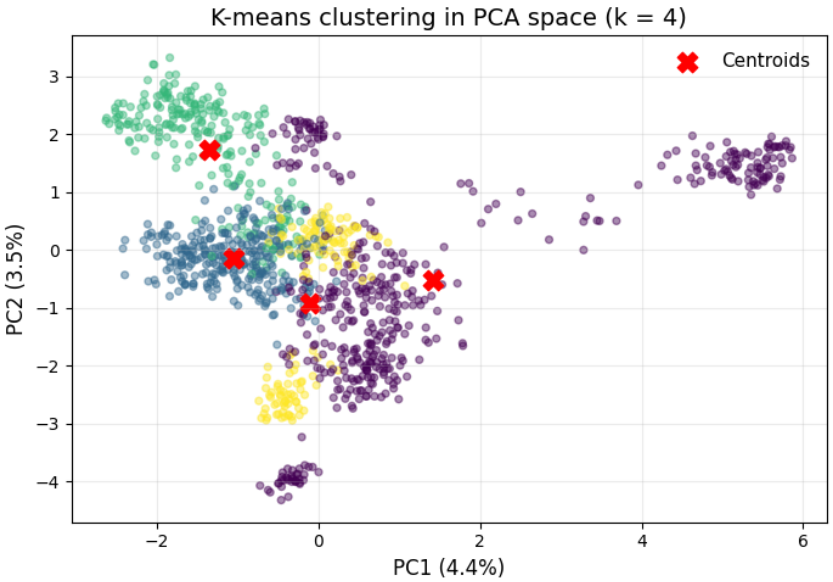
PCA captures the main directions of variability in the sequence data, but functional and non-functional proteins remain strongly overlapping, suggesting that function is not associated with a simple low-dimensional structure.

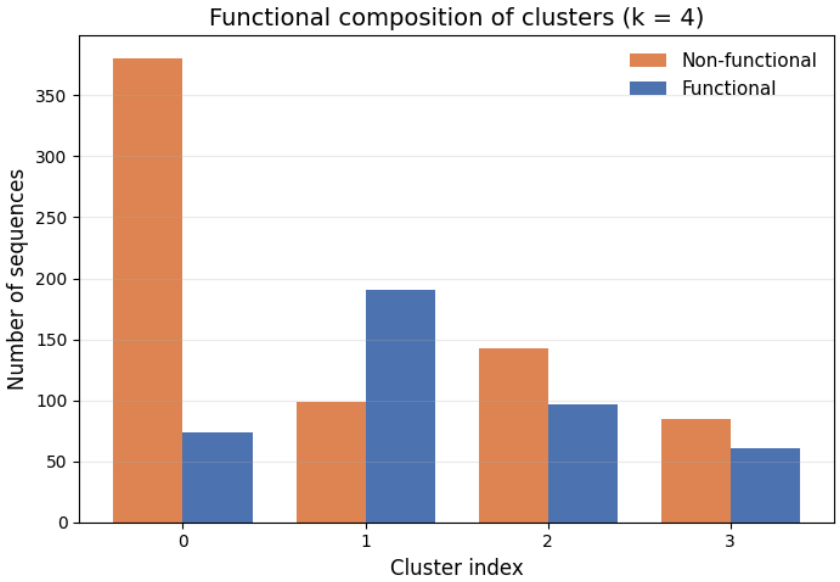
PCA: artificial sequences in natural space



Artificial sequences are projected onto the PCA space learned from natural data. They largely overlap with natural sequences, indicating that the global geometry of sequence space is preserved.

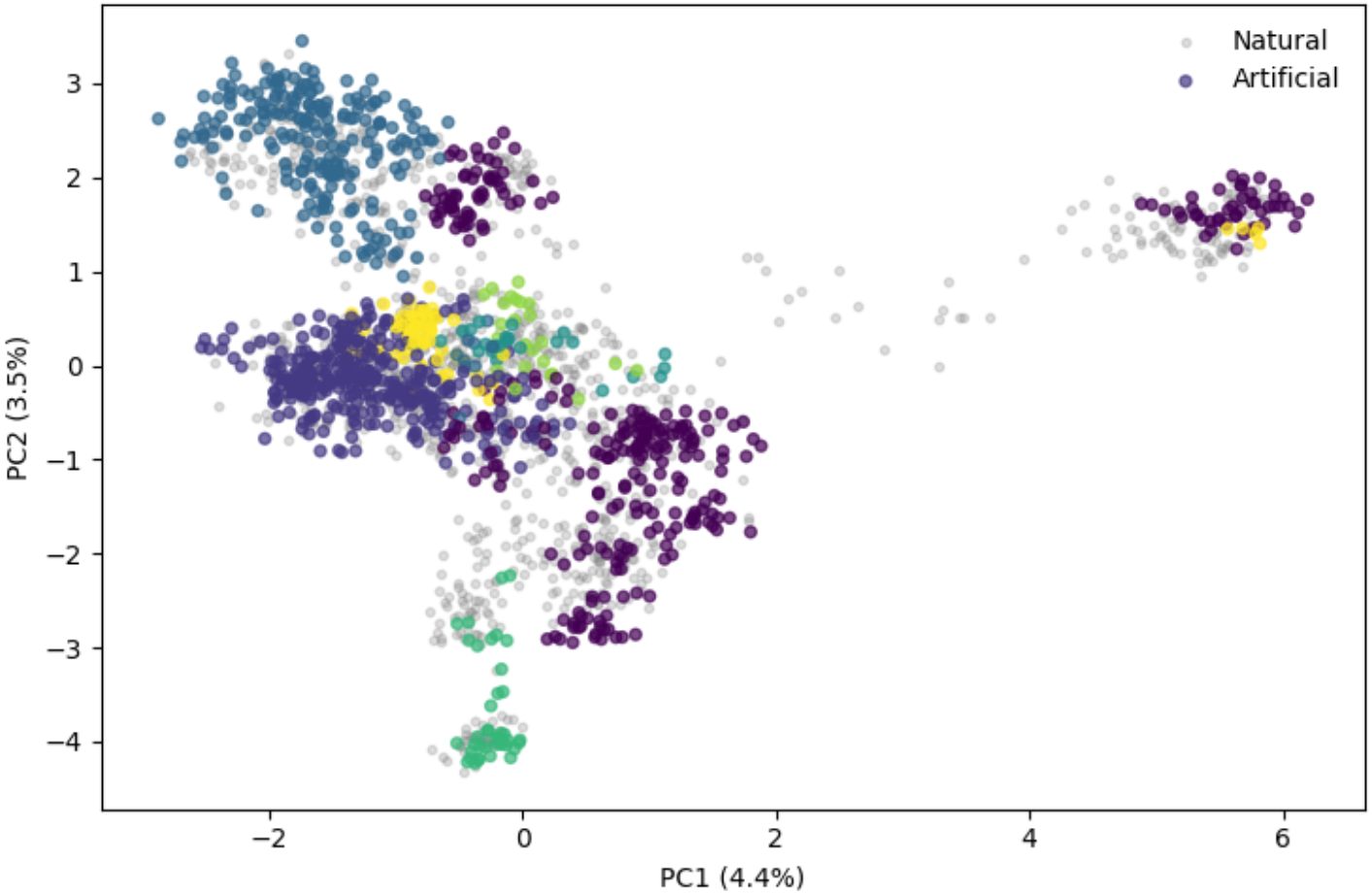
Unsupervised clustering: K-means





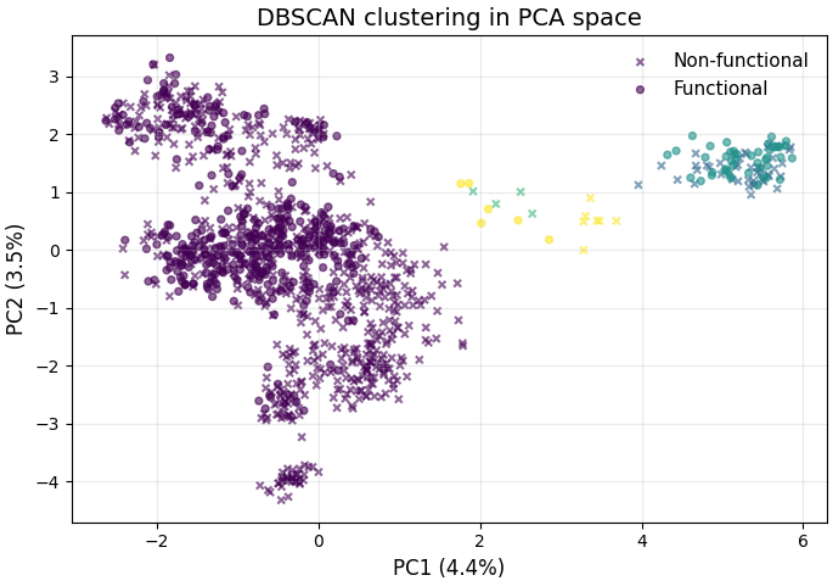
K-means identifies geometric structure in the PCA-projected sequence space, but cluster membership is not associated with protein functionality. Clusters contain a comparable mix of functional and non-functional sequences.

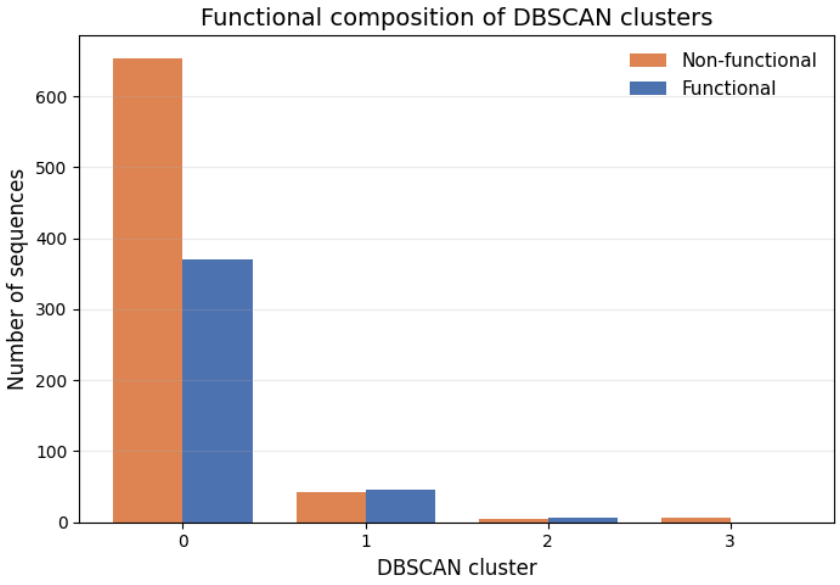
Clustering: artificial sequences vs natural clusters



Artificial sequences fall into the same geometric clusters as natural ones, instead of forming isolated regions.

Density-based clustering: DBSCAN





DBSCAN identifies dense regions in the projected space without fixing the number of clusters. Most sequences belong to a single dominant cluster, and functional and non-functional sequences remain strongly mixed across all groups.

Can function be predicted?

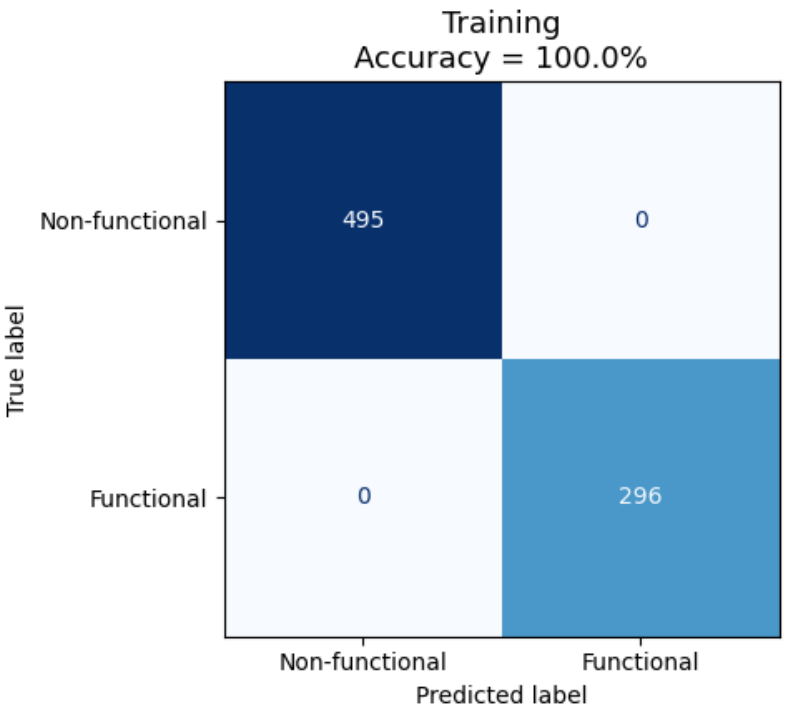
- **Input:** protein sequences encoded as high-dimensional vectors
- **Outputs:** functional vs non-functional
- **Goal:** predict function using supervised learning

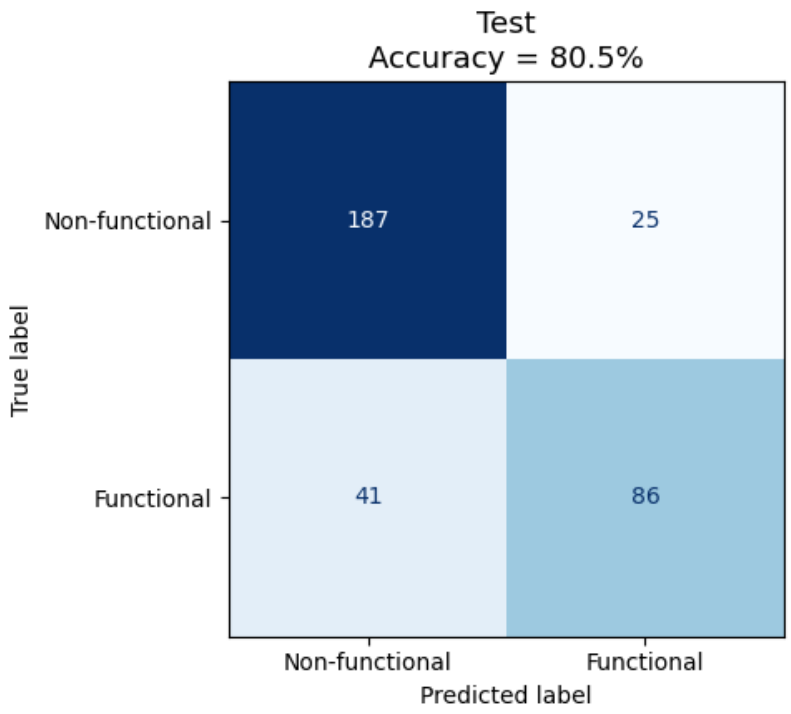
Key question

Can protein function be inferred from sequence information, even in the absence of clear unsupervised structure?

Logistic Regression

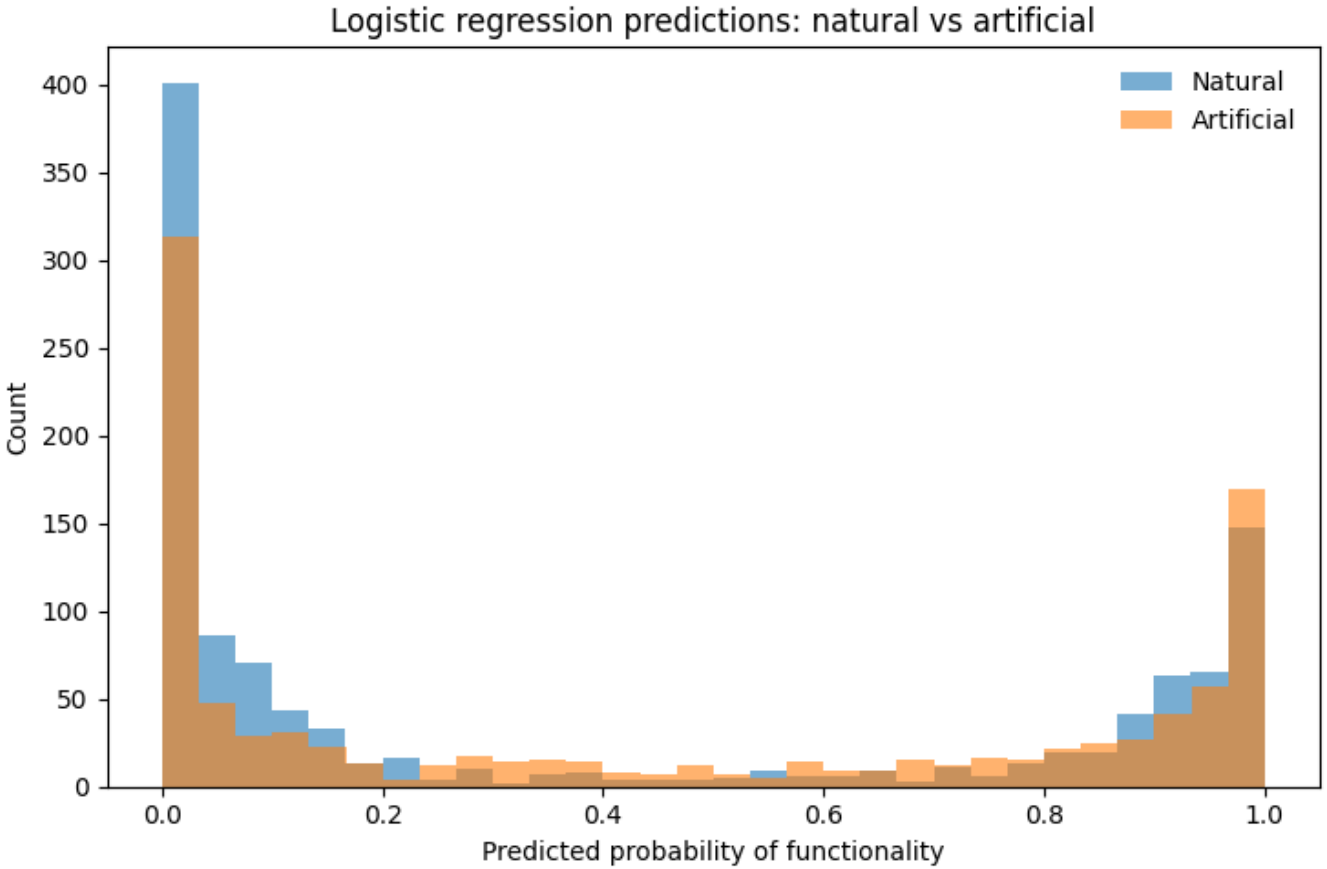
$$P(y = 1 \mid x) = \sigma(w^\top x + b), \qquad \sigma(z) = \frac{1}{1 + e^{-z}}$$





Good performance without explicit sequence structure.

Logistic Regression on artificial sequences



The logistic regression model trained on natural sequences is applied unchanged to artificial data. The distribution of predicted functional probabilities closely matches that of natural sequences, indicating that generated data are compatible with the decision boundary learned from real proteins.

Learning a generative model

- Previous tasks: prediction of function from sequence
- Here: learning the **distribution of sequences**

Objective

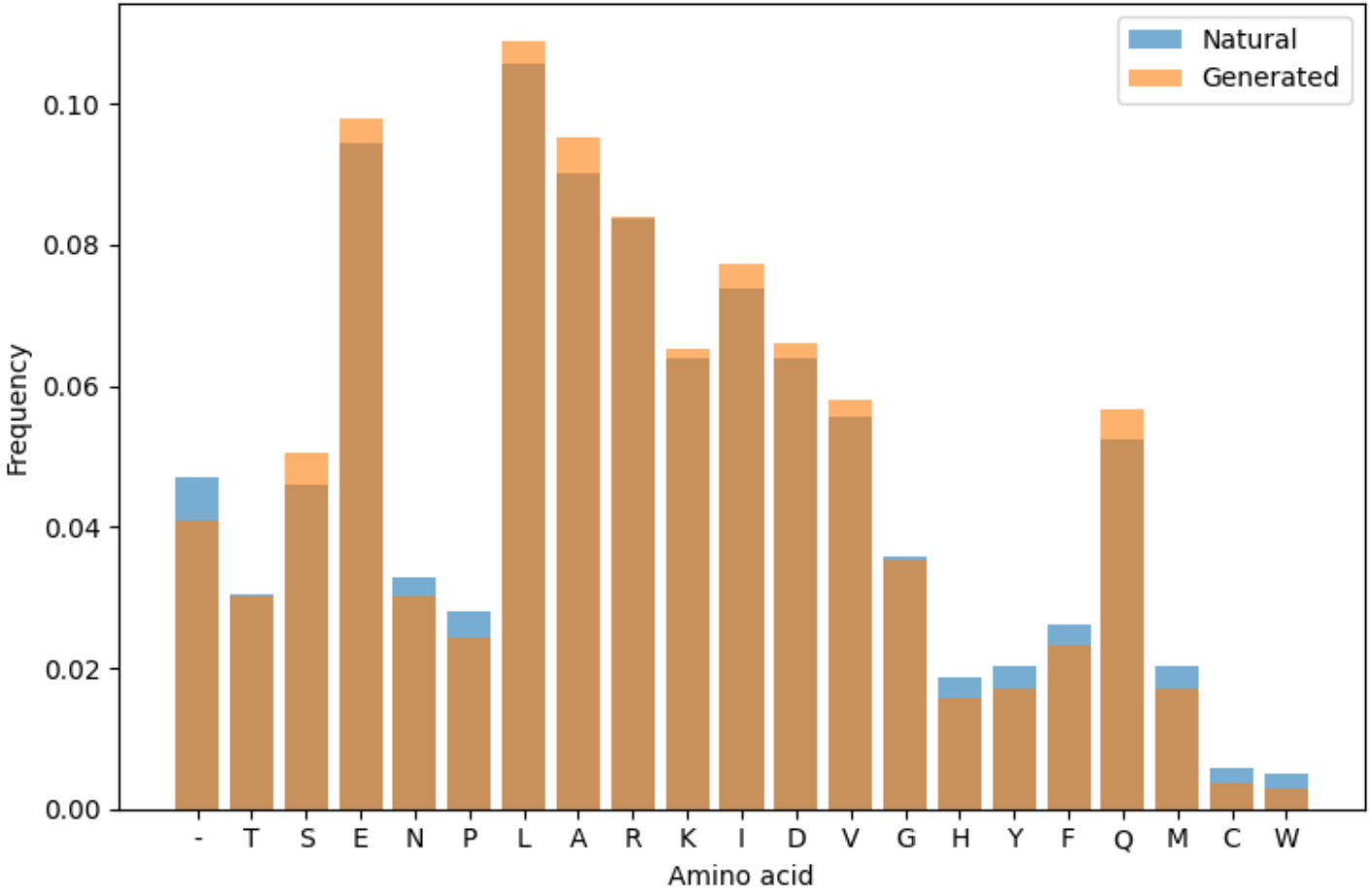
Model protein sequences in a low-dimensional latent space and generate new sequences consistent with the training data.

Learning a generative model

- **Goal:** learn a compact representation of protein sequence space
- **Encoder:** maps sequences to a low-dimensional latent space
- **Latent space:** continuous, structured, regularized
- **Decoder:** generates new sequences by sampling from latent space

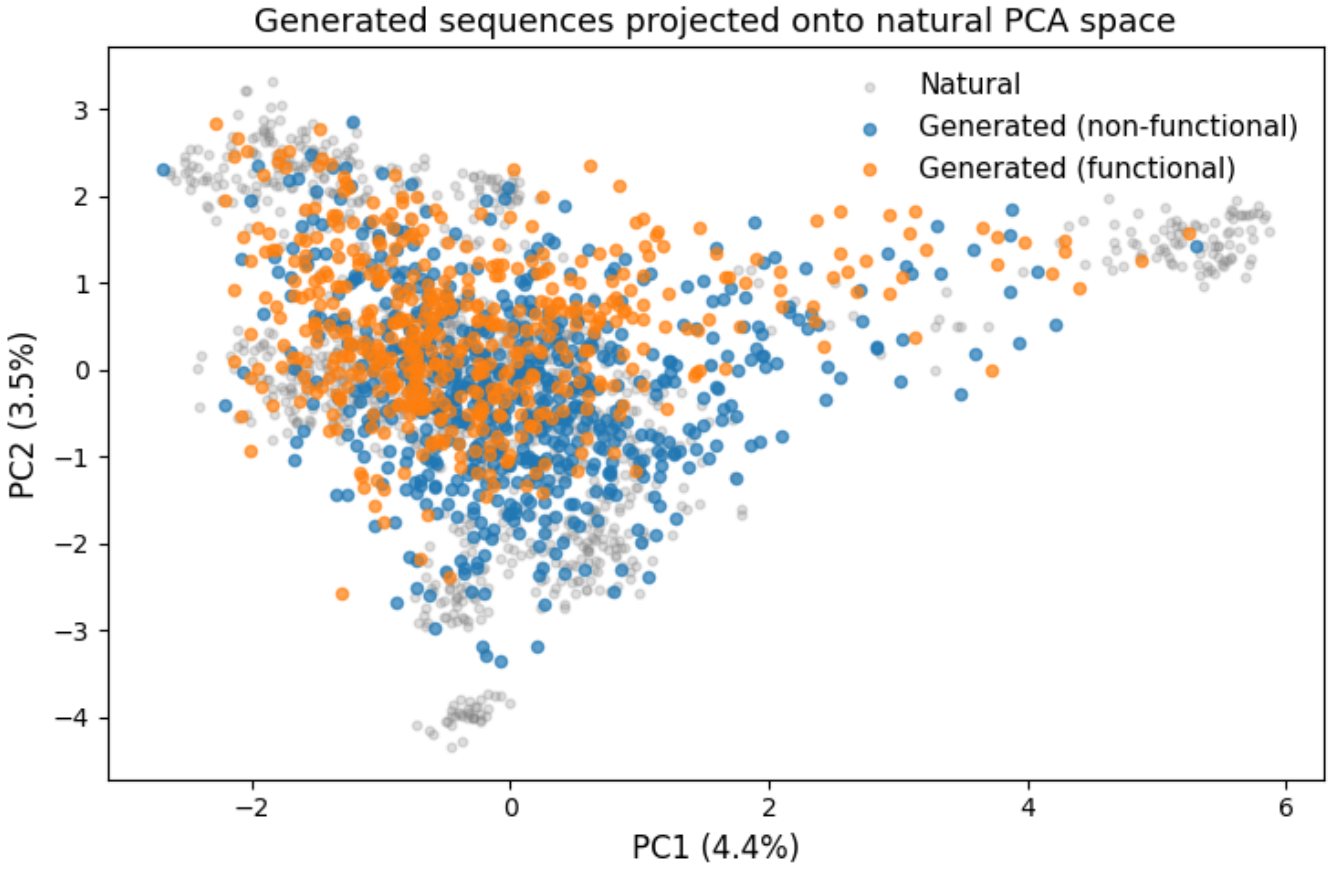
$$\text{sequence } x \;\rightarrow\; z \sim \mathcal{N}(0, I) \;\rightarrow\; \text{new sequence}$$

Amino acid distribution: natural vs generated



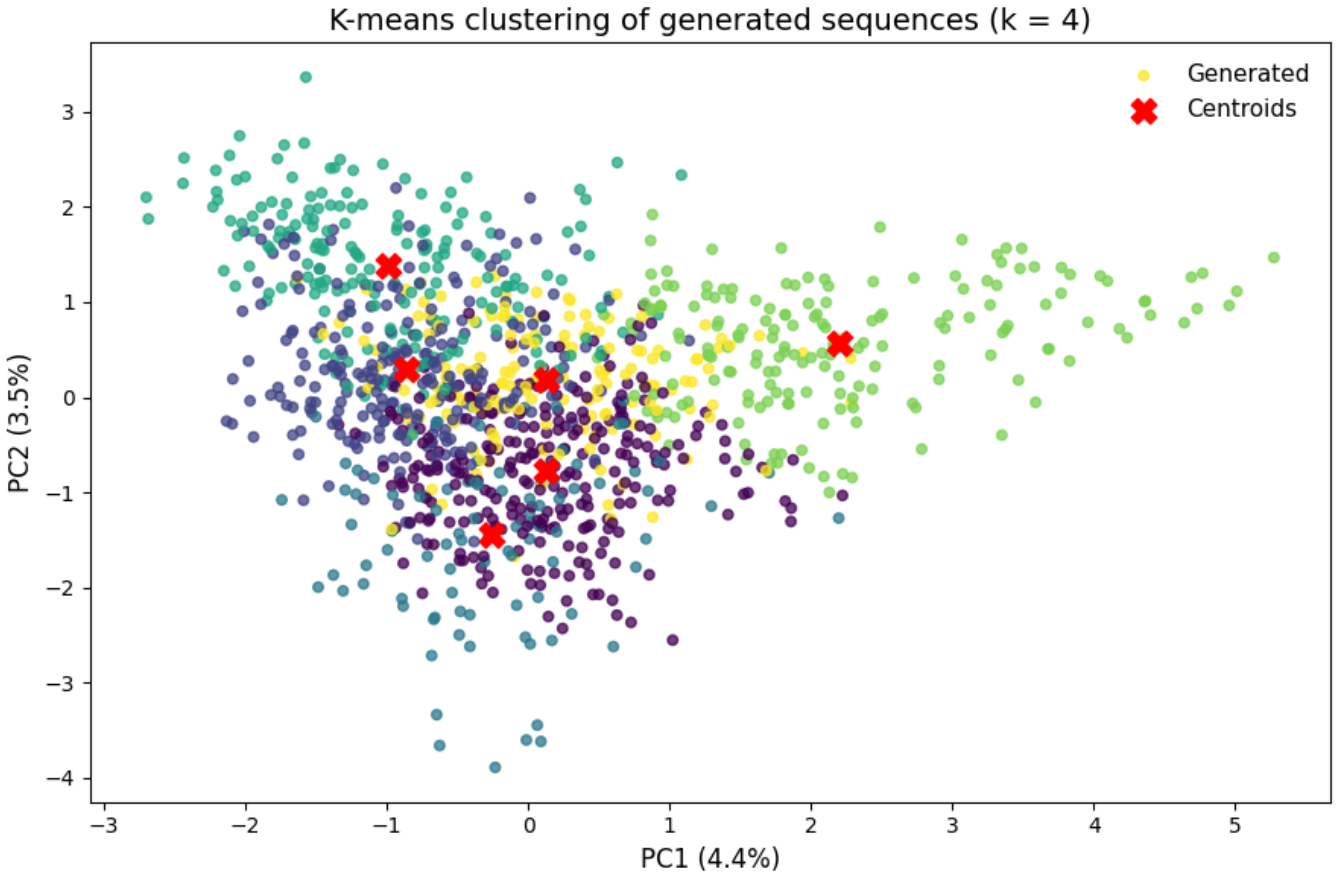
Generated sequences reproduce the marginal amino acid frequencies of natural proteins, indicating that the VAE captures basic compositional statistics.

PCA: natural vs generated sequences



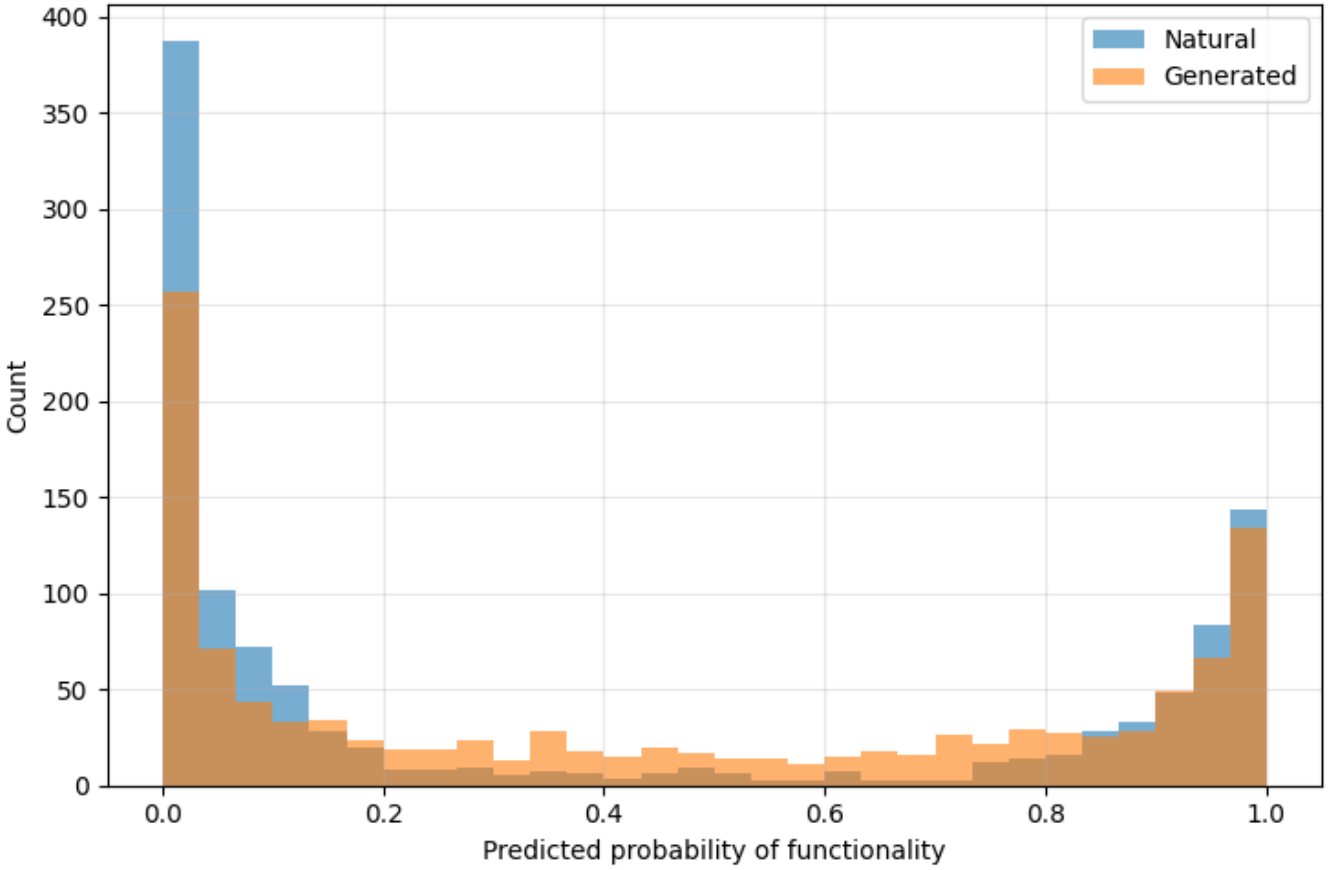
Generated sequences largely overlap with natural ones in PCA space, indicating that the VAE captures the global structure of sequence space, while slightly exploring lower-density regions.

Clustering of generated sequences



Generated sequences are clustered with k-means and shown in the PCA space defined by natural sequences.

Logistic Regression on generated sequences



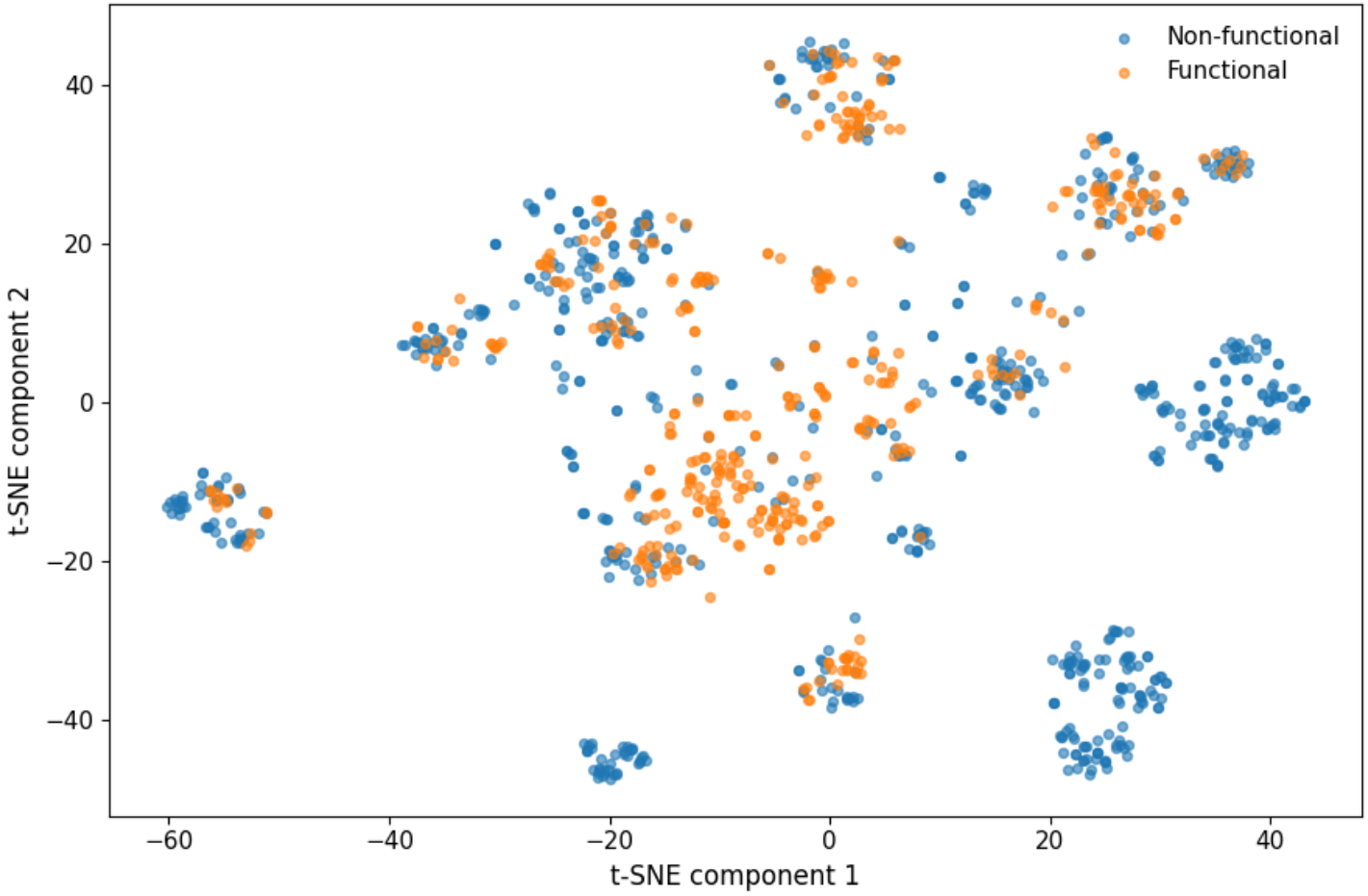
Distribution of predicted functional probability for natural and generated sequences.

Take-home message

- Protein sequences form a high-dimensional space where function is not clearly separated by unsupervised methods.
- Despite this, supervised models can learn predictive signals directly from sequence data.
- A variational autoencoder captures the global structure of sequence space and can generate new sequences.
- Generated sequences closely resemble natural ones under several independent comparisons.

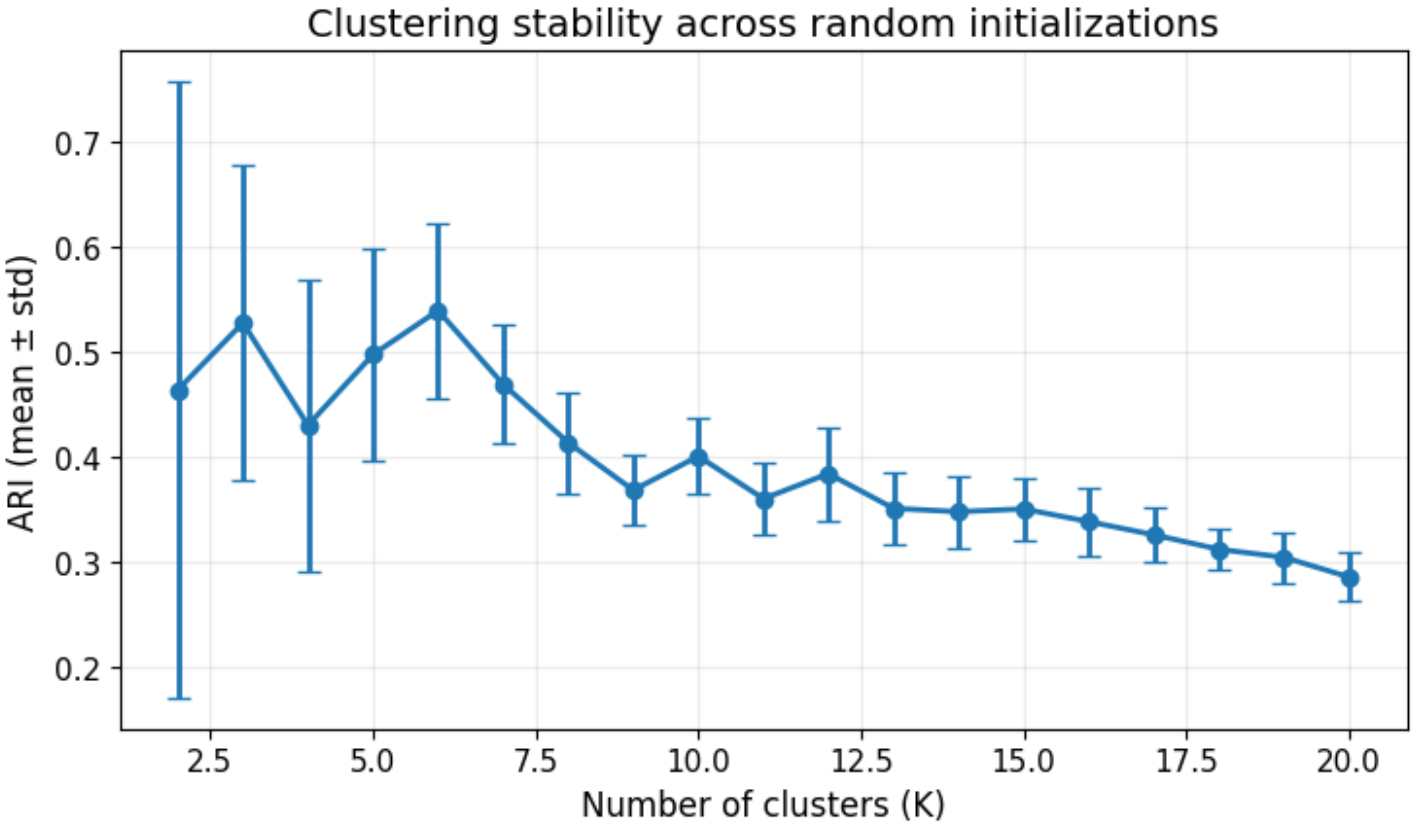
Thank you for your attention!

Bakcup: non-linear dimensionality reduction: t-SNE



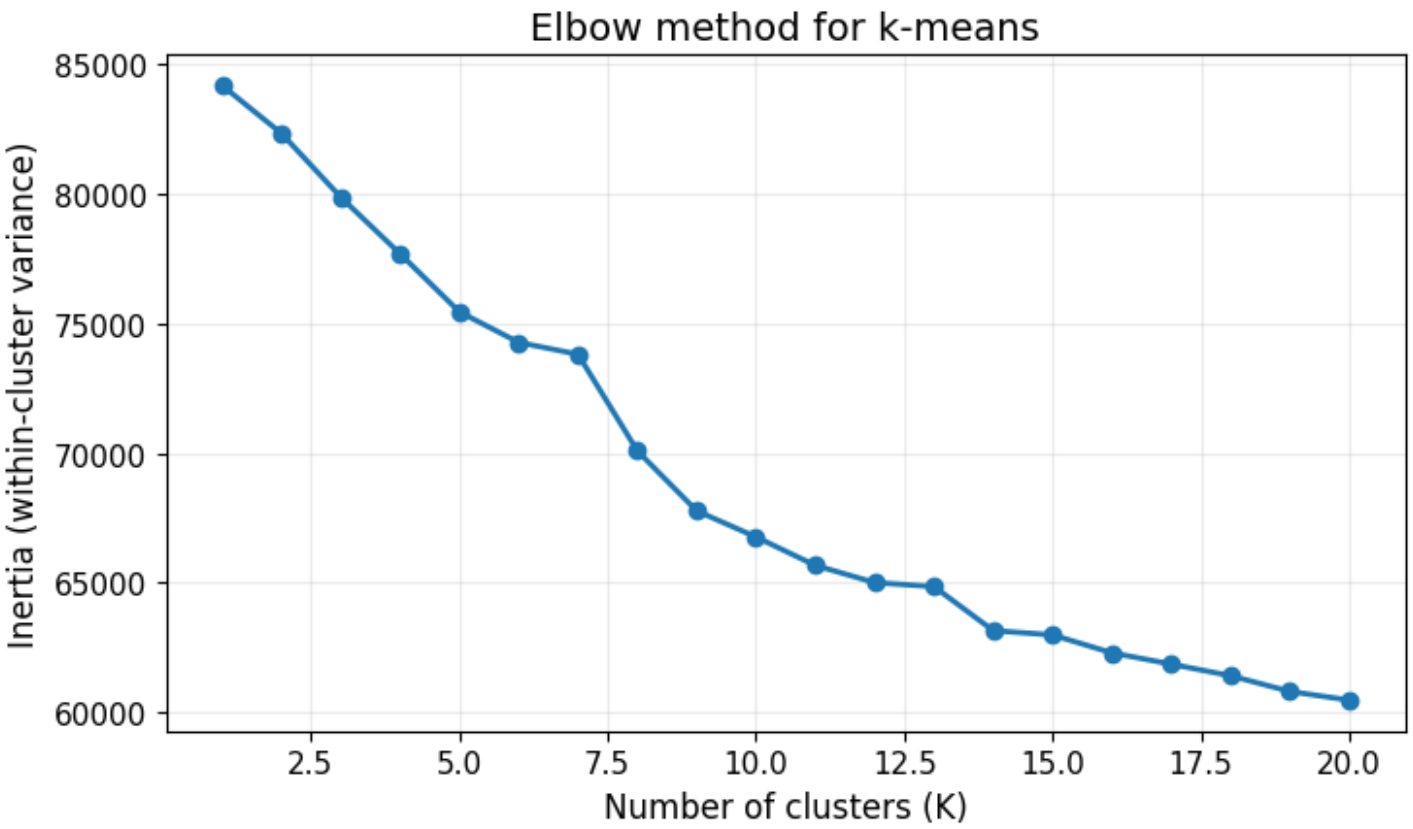
t-SNE emphasizes local neighborhood structure. Despite revealing small-scale organization, functional and non-functional sequences remain largely mixed.

Backup: clustering stability across initializations



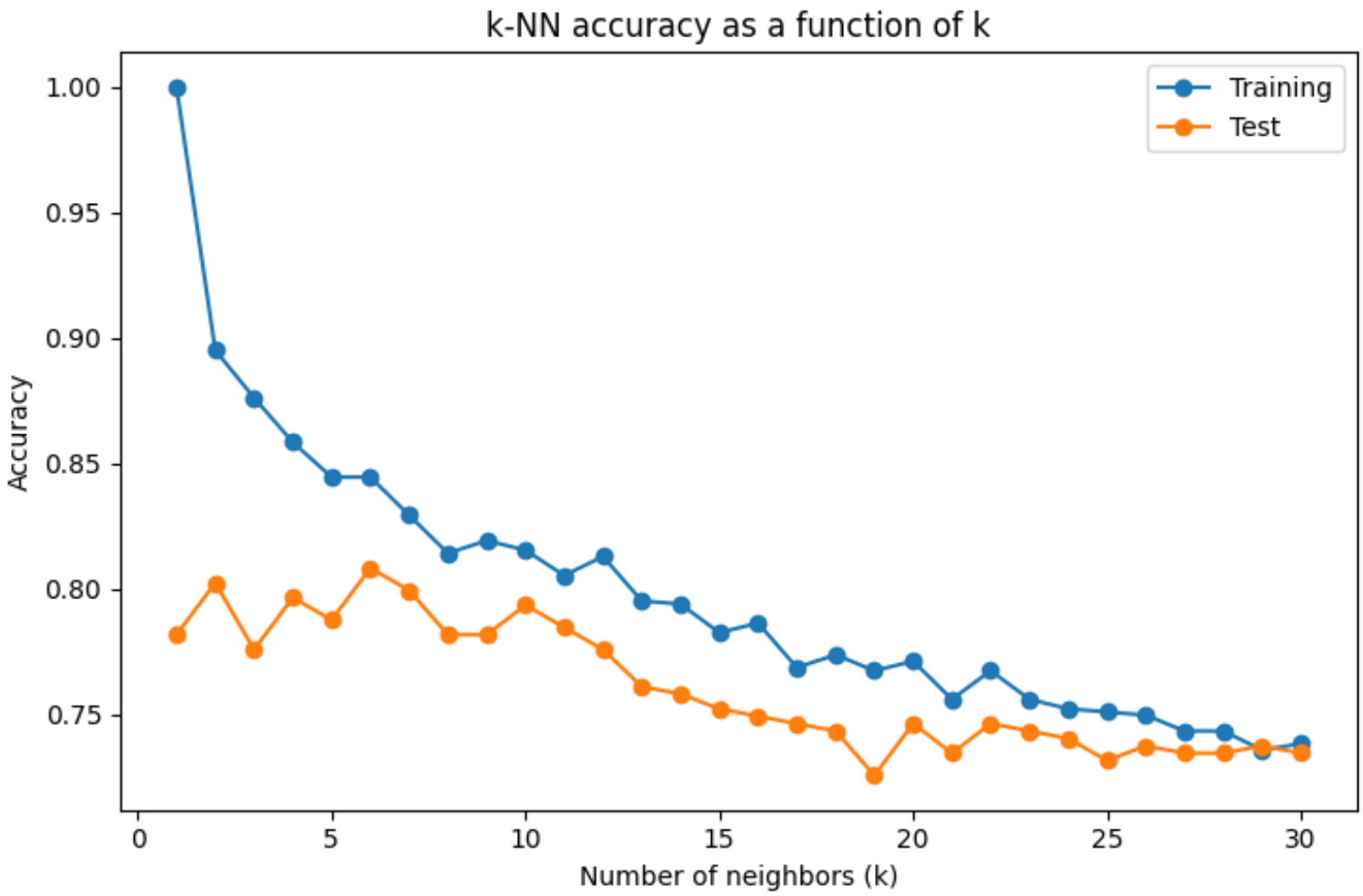
Adjusted Rand Index (ARI) computed across multiple random initializations to assess the robustness of k-means clustering. Higher ARI indicates more stable partitions.

Backup: choosing the number of clusters: Elbow method



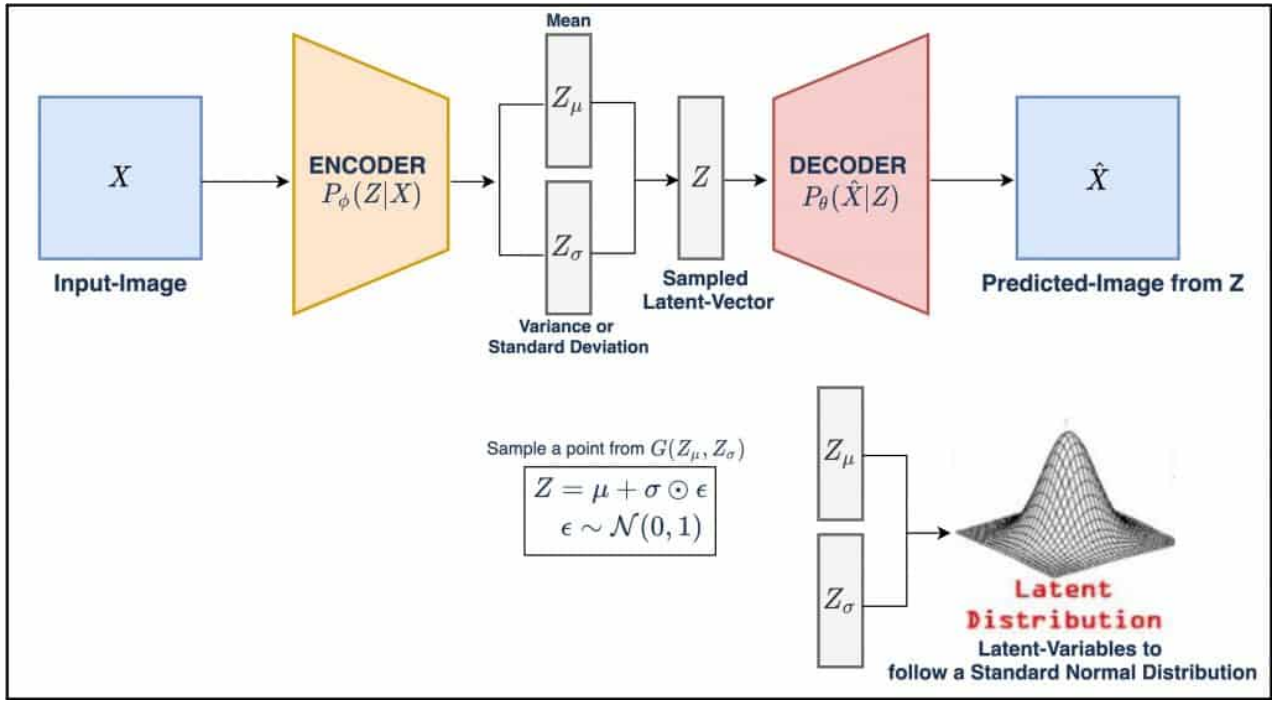
The inertia decreases smoothly with the number of clusters, without a clear elbow point, suggesting the absence of a natural cluster scale in the data.

Backup: k-NN classifier — choice of k



Training accuracy decreases with increasing k, while test accuracy shows a broad maximum for intermediate values. This reflects the bias–variance trade-off and motivates the choice of a moderate number of neighbors.

Backup: Variational Autoencoder (VAE) architecture



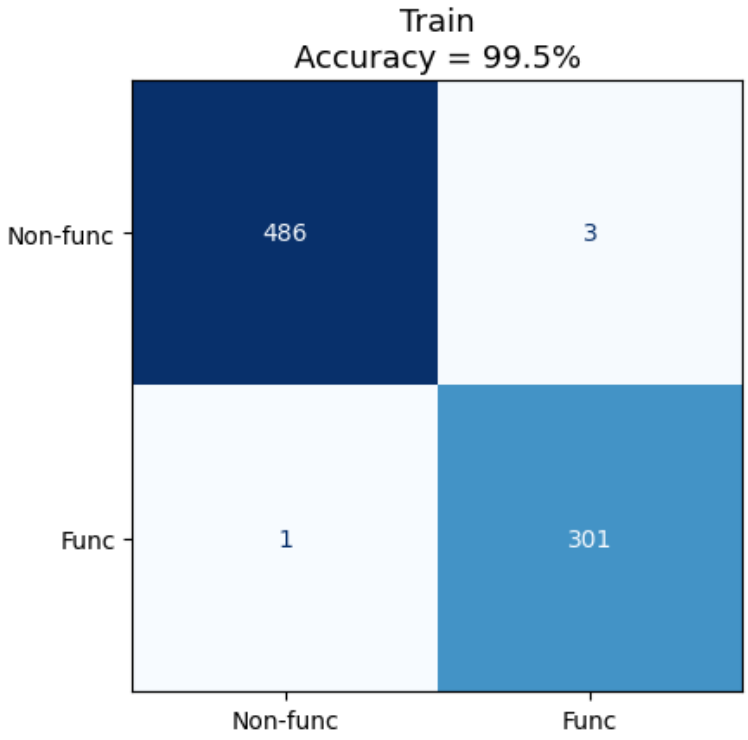
A VAE consists of an **encoder** that maps an input sequence x to a distribution in latent space (μ, σ), and a **decoder** that reconstructs sequences from samples

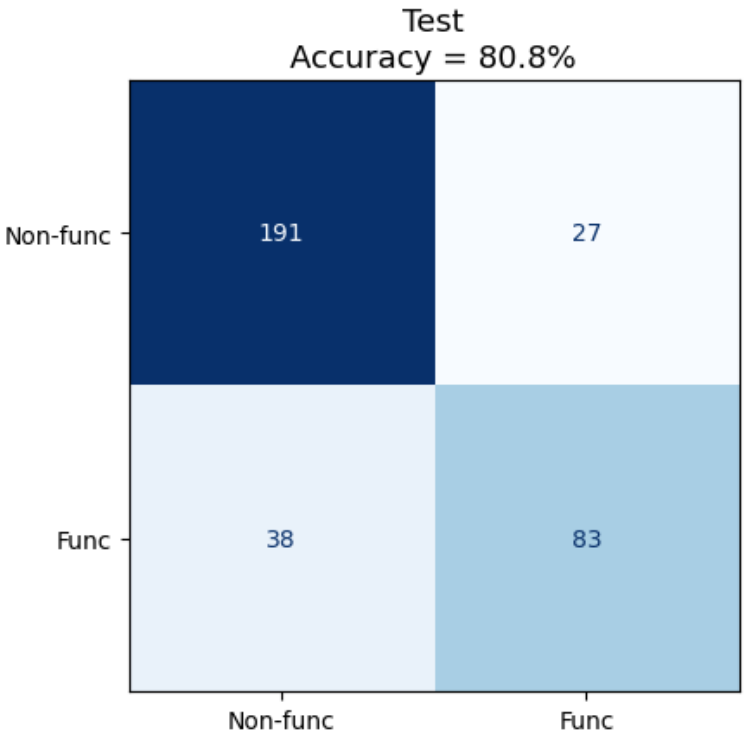
$$z_i = \mu_i + \sigma_i \varepsilon_i, \qquad \varepsilon \sim \mathcal{N}(0, I).$$

The latent space is regularized to follow a standard normal distribution, allowing smooth interpolation and generation of new sequences.

Backup: convolutional Neural Network

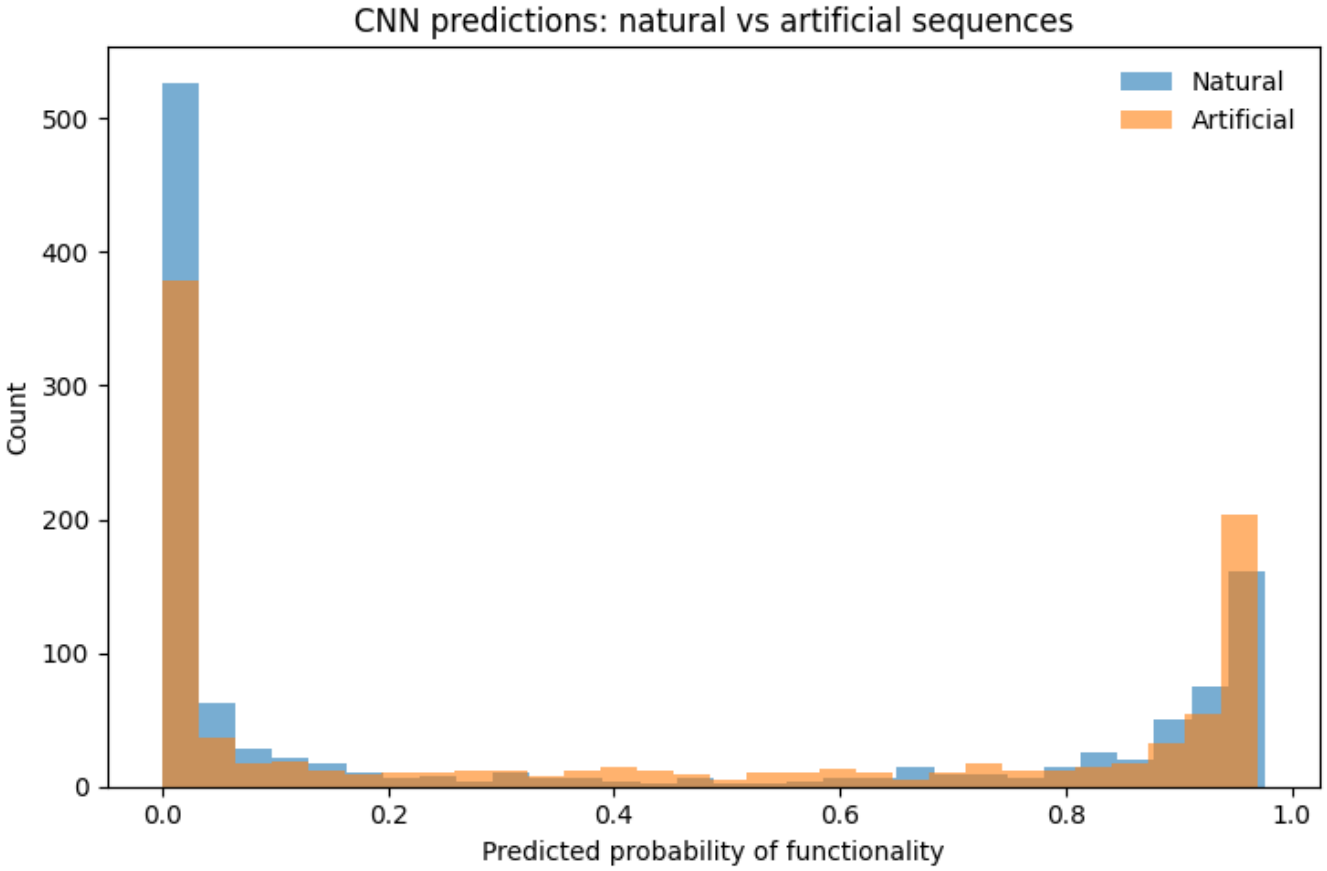
$$x \in \mathbb{R}^{20 \times L} \xrightarrow{\text{1D convolutions}} \text{local motifs} \xrightarrow{\text{pooling + FC}} P(y = 1 \mid x)$$





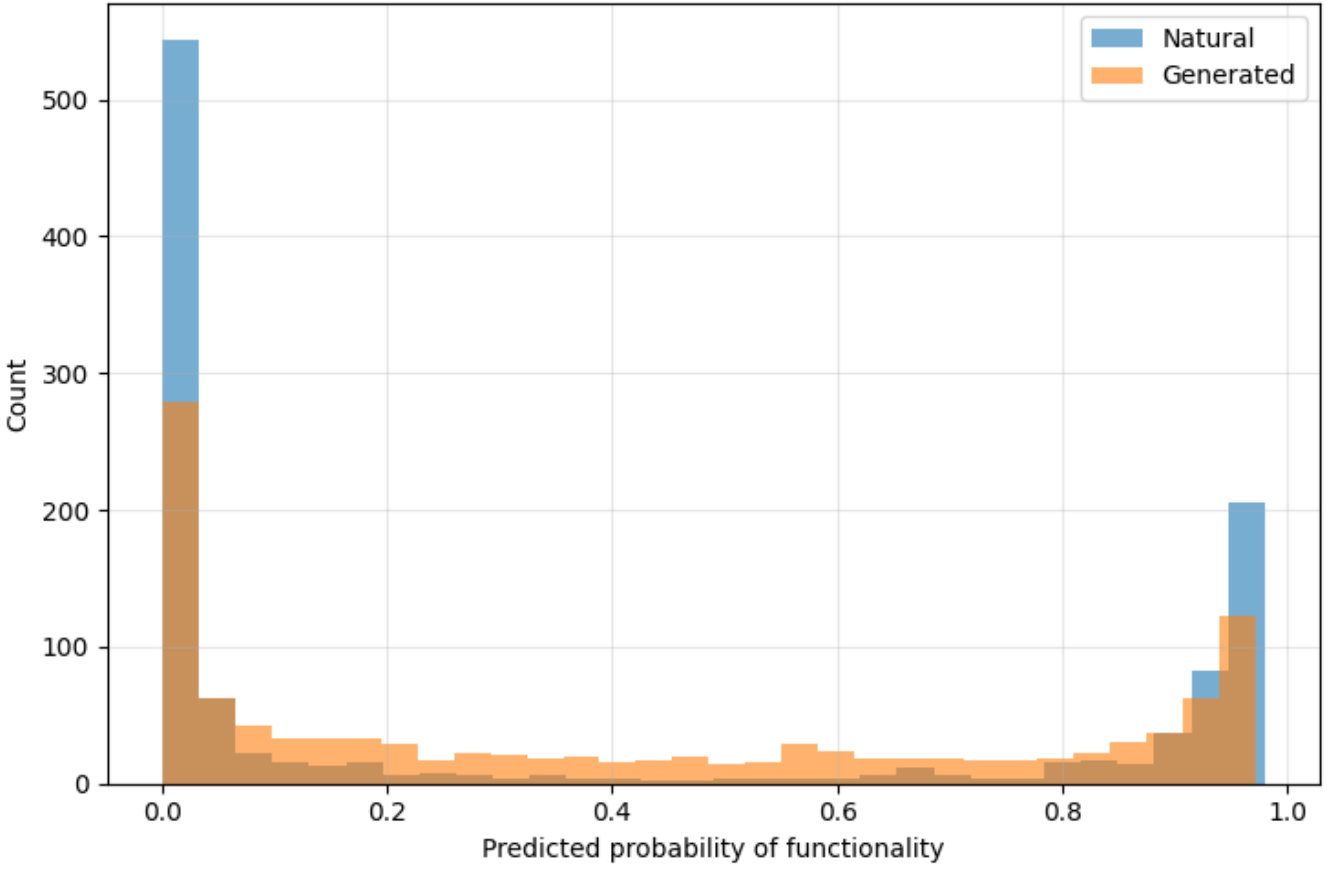
Non-linear model capturing local sequence patterns, but test performance comparable to linear baselines.

Backup: CNN on artificial sequences



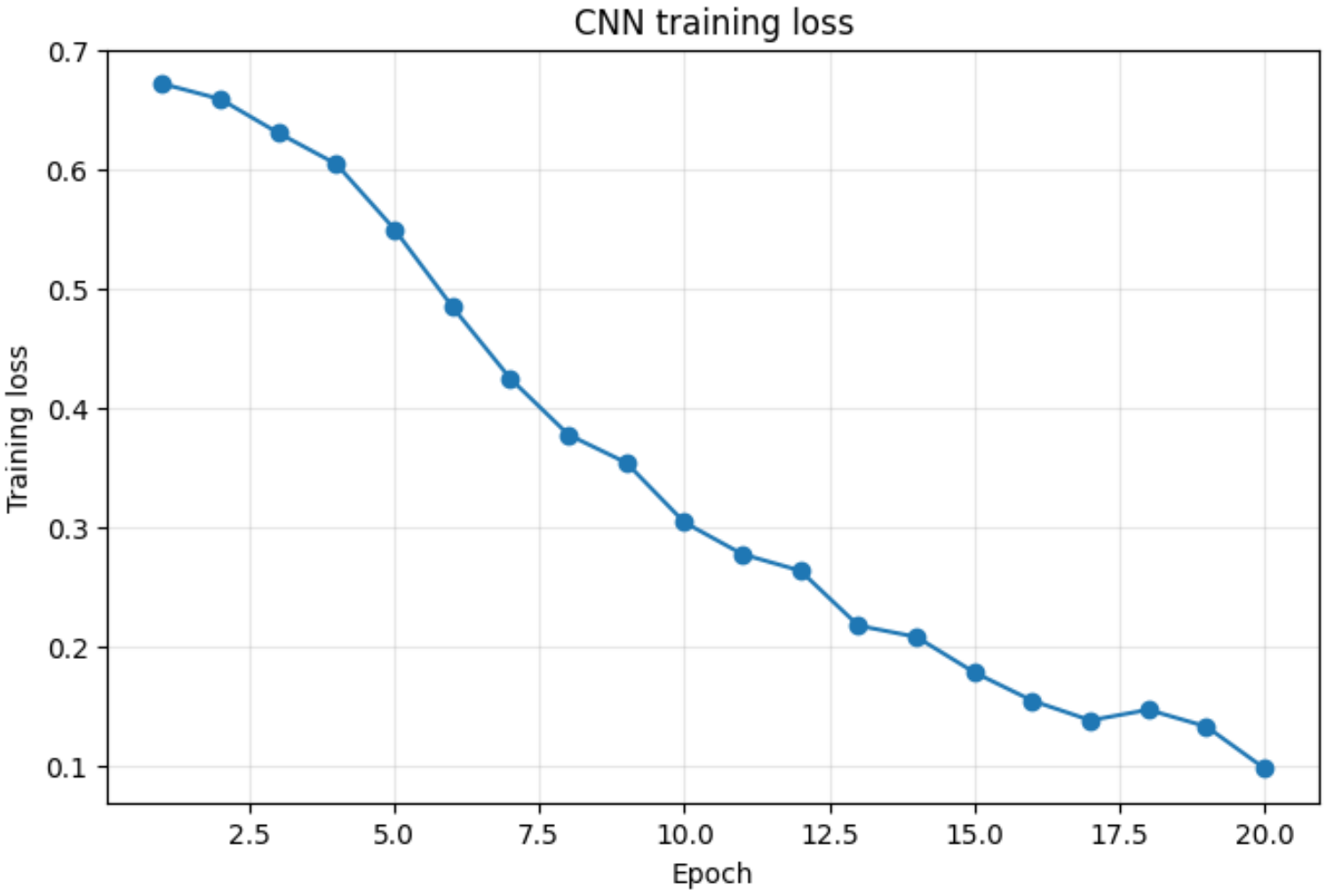
CNN predictions on artificial sequences show a smooth probability distribution, comparable to that of natural data. This suggests that generated sequences do not exploit spurious patterns and remain consistent with local sequence features learned by the network.

Backup: CNN on generated sequences



CNN predictions on generated sequences show a probability distribution comparable to natural data.

Backup: CNN training dynamics



The training loss decreases smoothly over epochs, indicating stable optimization. No signs of divergence or instability are observed during training.